

REPORT

on

Post Lab

Submitted in partial fulfilment

of Subject

Image Processing Lab (N-PECCS502P)

(V Semester, B.Tech. CSE)

(ACADEMIC SESSION 2026-27)

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**(AN AUTONOMOUS INSTITUTION AFFILIATED TO RASHTRASANT TUKADOJI MAHARAJ
NAGPUR UNIVERSITY, NAAC ACCREDITED WITH 'A' GRADE)**

POST LAB DETAILS

USN No.	CS25D002
STUDENT NAME	Avni A. Meshram
POST LAB	Create a website for Image Processing techniques. (Include all 12 practical's)
GitHub REPOSITORY URL:	https://github.com/2006avni/Image-Processing
GitHub REPOSITORY URL QR CODE:	
DEPLOYMENT URL:	
DEPLOYMENT URL QR CODE:	

1. ABSTRACT

The Image Processing Lab is a web-based application developed to demonstrate various image processing operations using Python, OpenCV, HTML, CSS, and JavaScript. The system allows users to upload an image and apply operations such as grayscale conversion, arithmetic and bitwise operations, geometric transformations, image enhancement, spatial filtering, inpainting, compression, morphological processing, and correlation-based object detection. A Flask backend processes the uploaded images using OpenCV and displays the original and processed images for easy comparison. The application provides an interactive and user-friendly platform for understanding image processing concepts and observing their practical results. This project reduces the complexity of performing image processing experiments and helps students learn the implementation and applications of computer vision techniques.

2. INTRODUCTION

Image processing is an important field of computer science that focuses on manipulating and improving digital images using computational techniques. It is widely used in medical imaging, satellite imaging, security systems, photography, robotics, pattern recognition, and computer vision. Image processing helps in enhancing image quality, removing noise, detecting objects, extracting useful information, and preparing images for further analysis.

The Image Processing Lab is a web-based application designed to demonstrate fundamental image processing operations using Python, OpenCV, Flask, HTML, CSS, and JavaScript. The application provides an interactive interface where users can upload an image, select a practical operation, and observe the processed output. It displays the original and processed images side by side, making it easier to understand the effect of each operation.

The project covers various image processing techniques, including RGB-to-grayscale conversion, arithmetic and bitwise operations, geometric transformations, image enhancement, spatial filtering, image inpainting, lossless compression, morphological operations, and correlation-based object detection. The Flask backend performs the actual processing using OpenCV, while the frontend provides a simple and attractive user interface.

3. TECHNOLOGY STACK

Frontend: HTML5, CSS, and JavaScript

Backend: Python with Flask

Image Processing: OpenCV

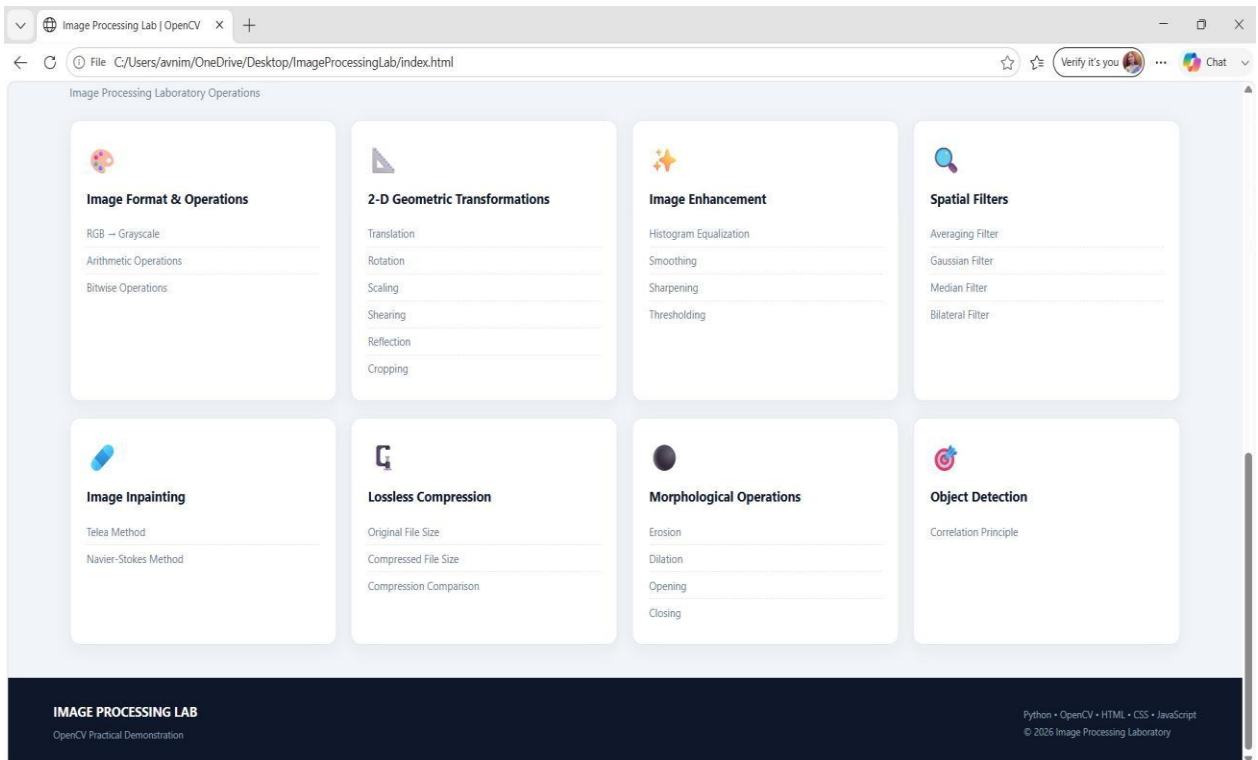
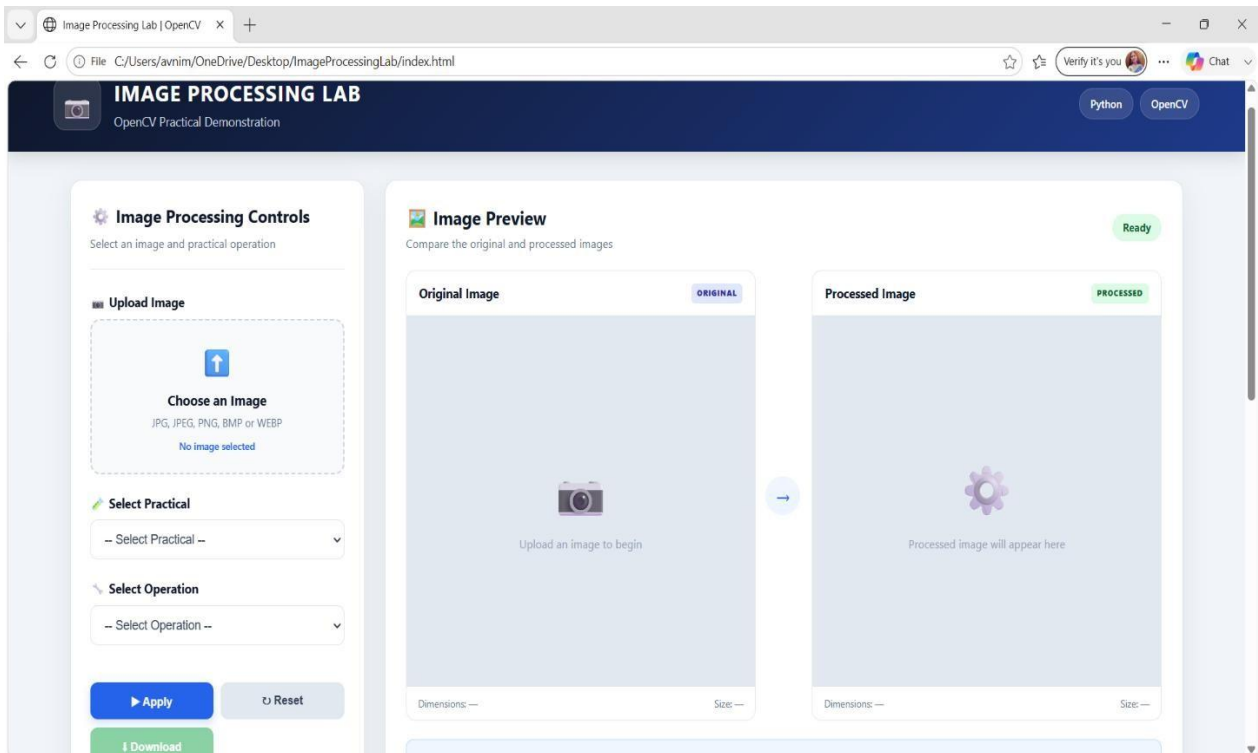
Numerical Operations: NumPy

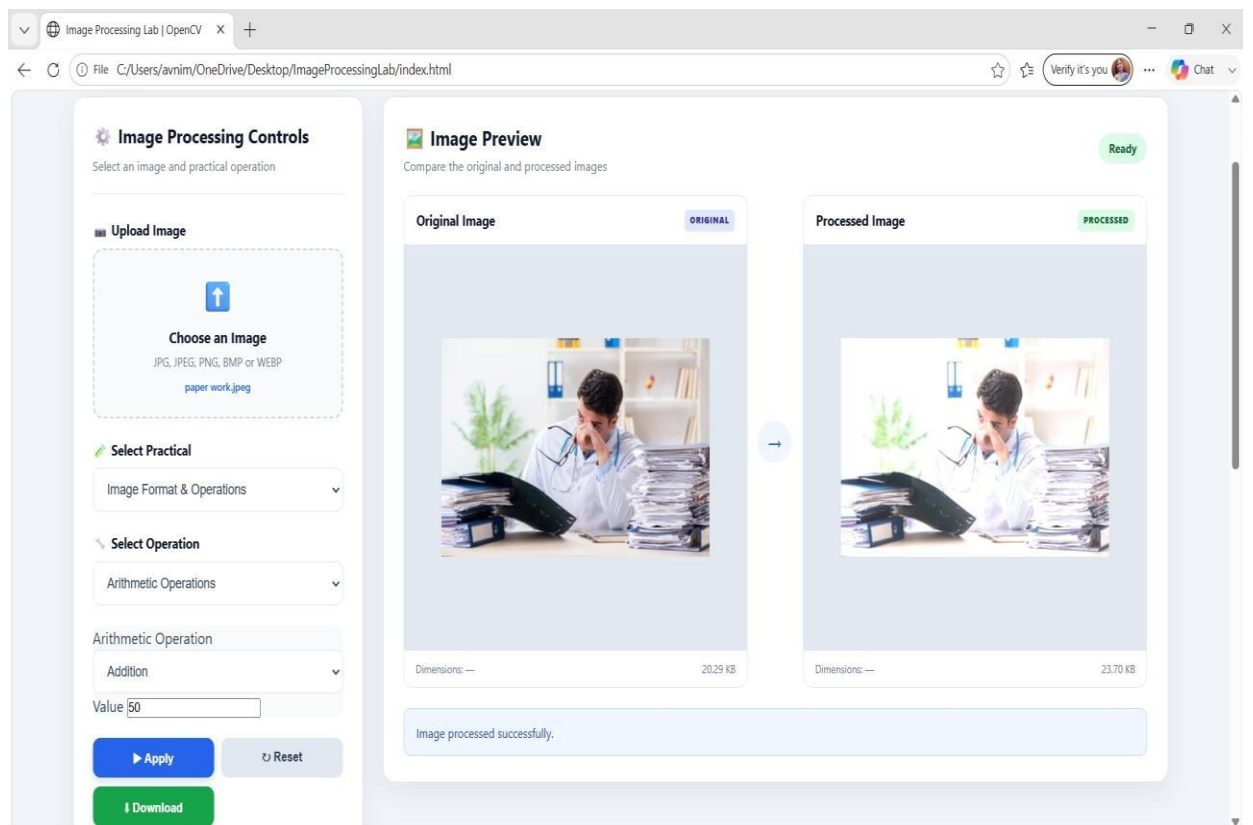
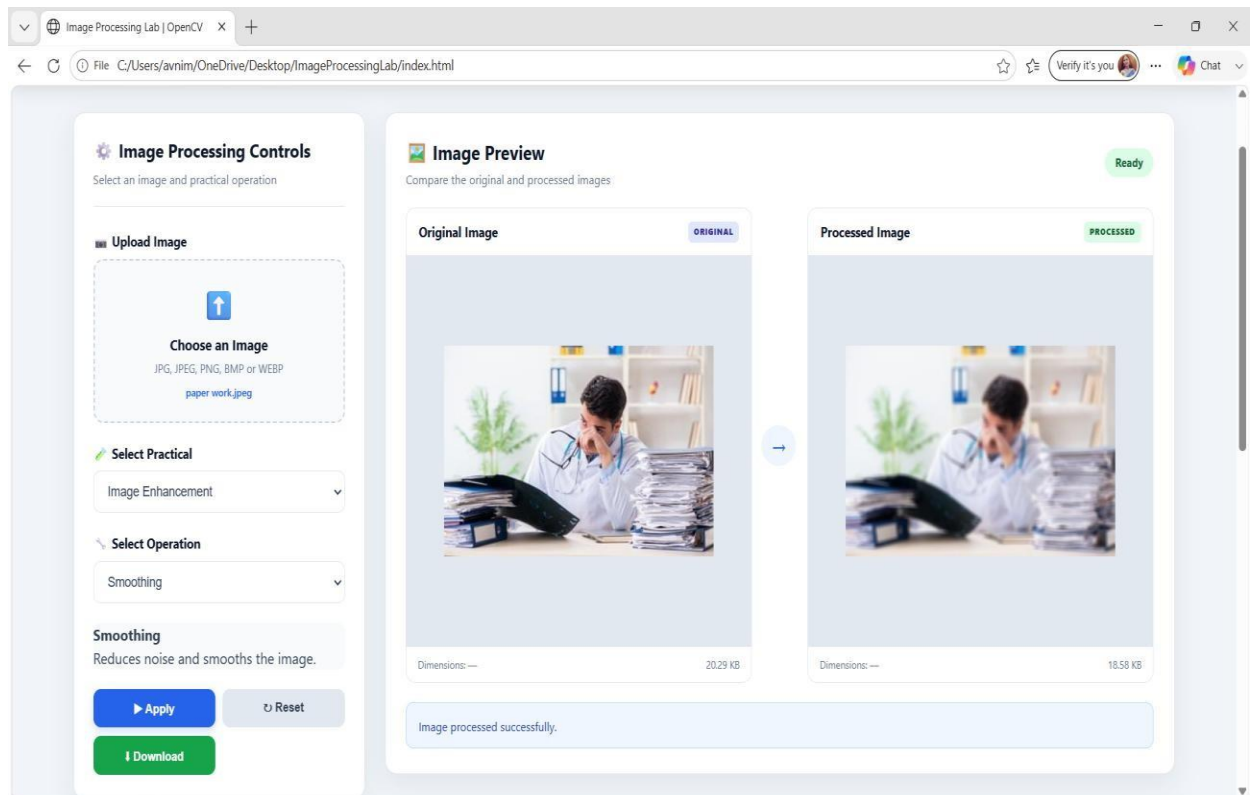
Image Handling: Pillow

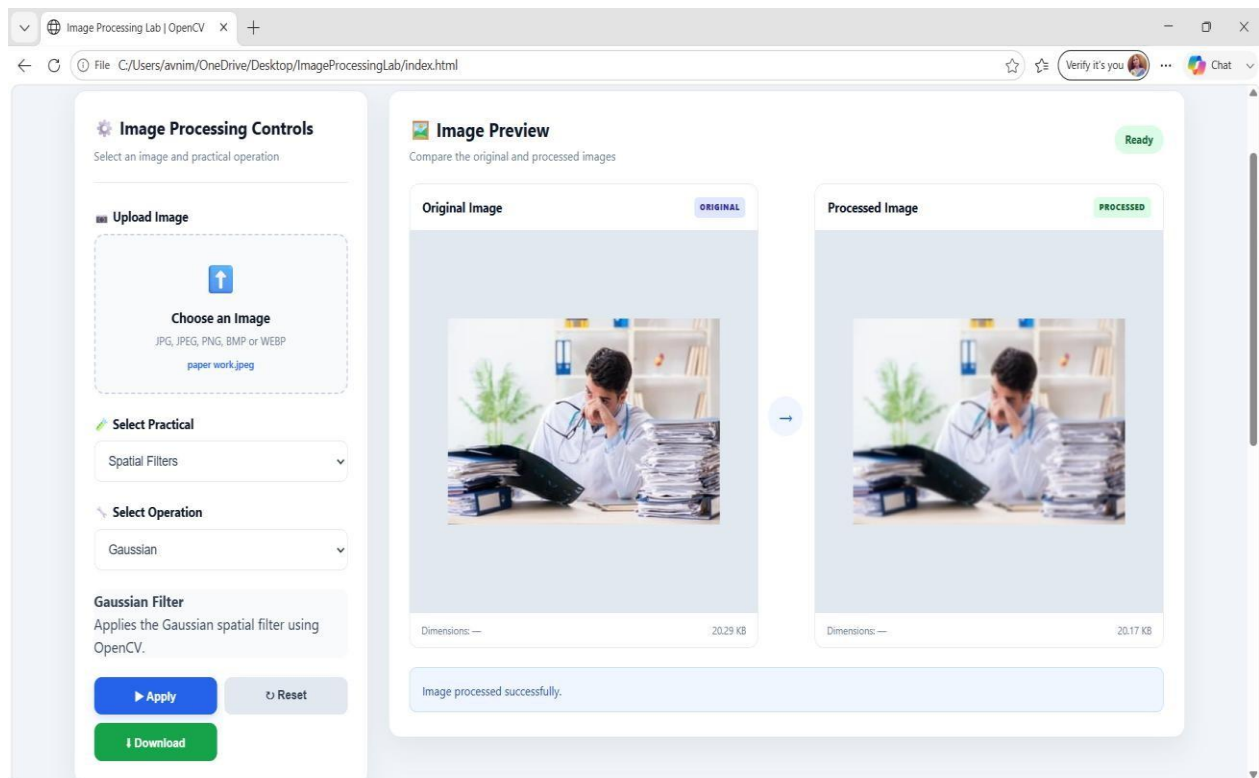
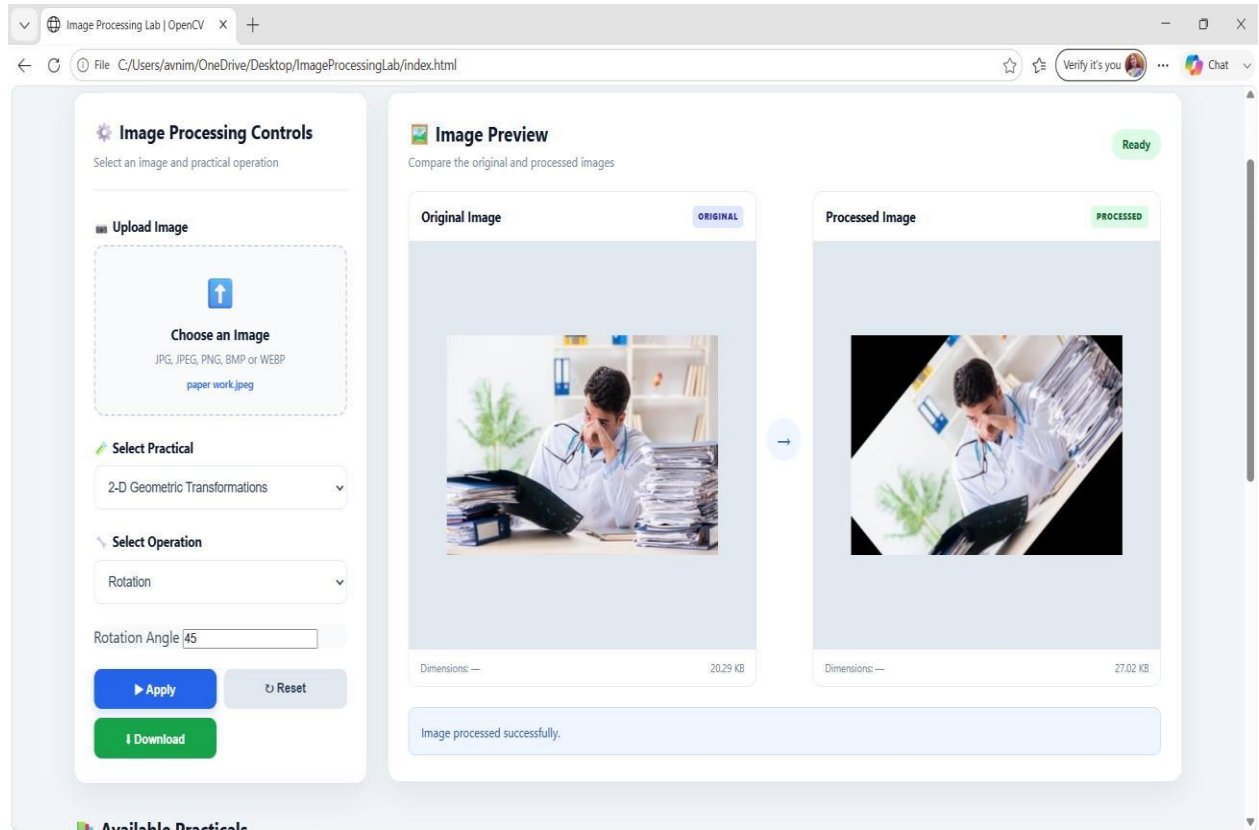
API: Flask REST API

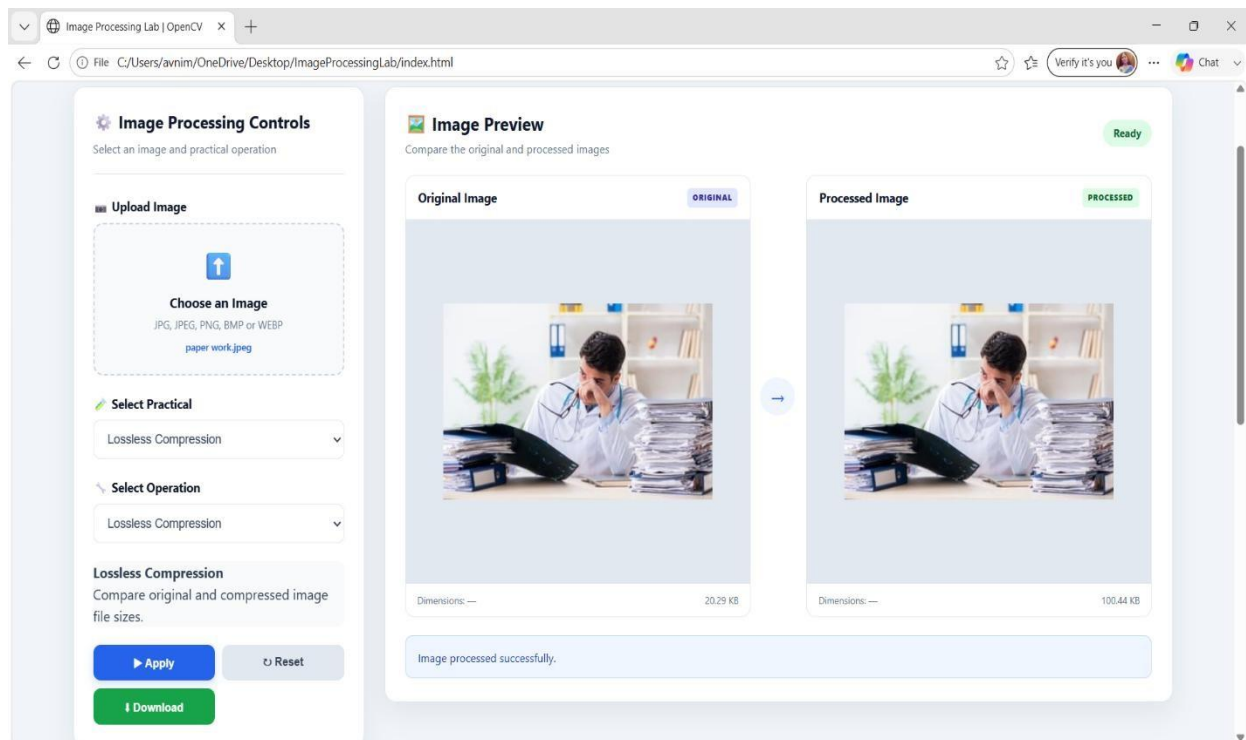
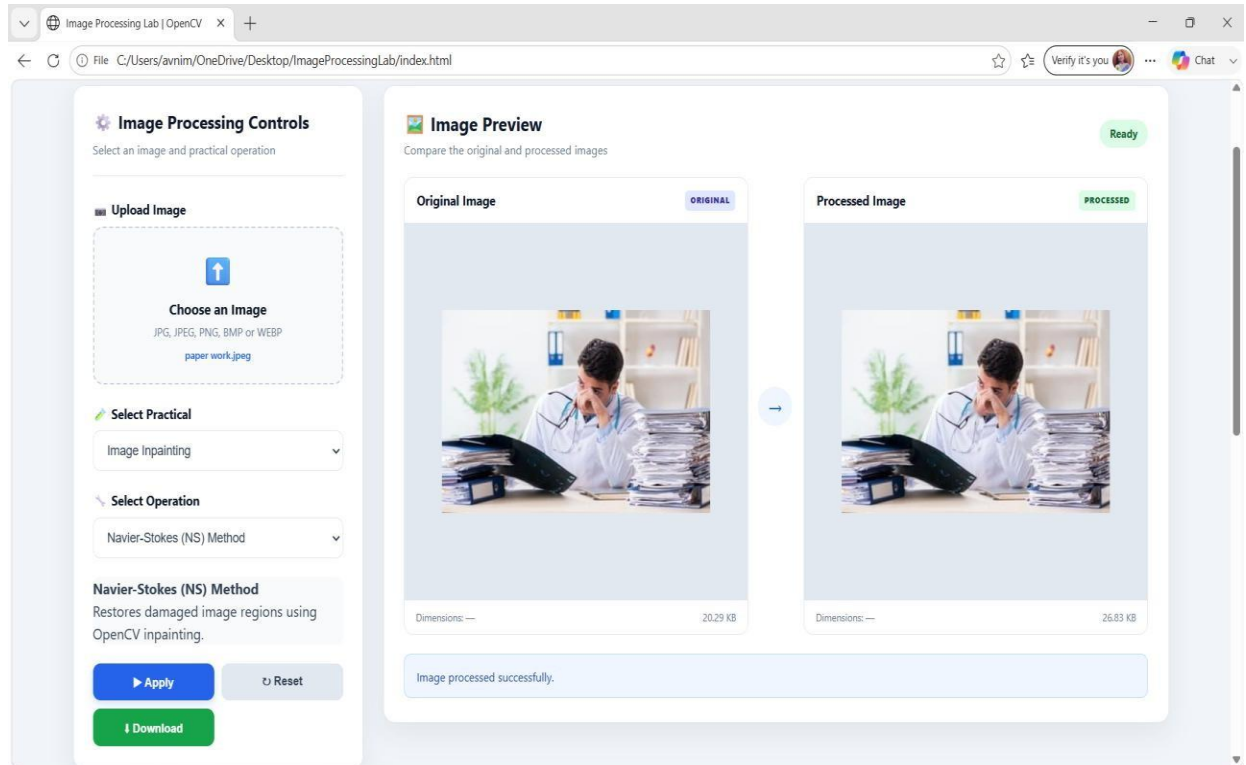
Database: Not required; the project uses temporary local file storage for uploaded and processed images.

4. APPLICATION SCREENSHOTS









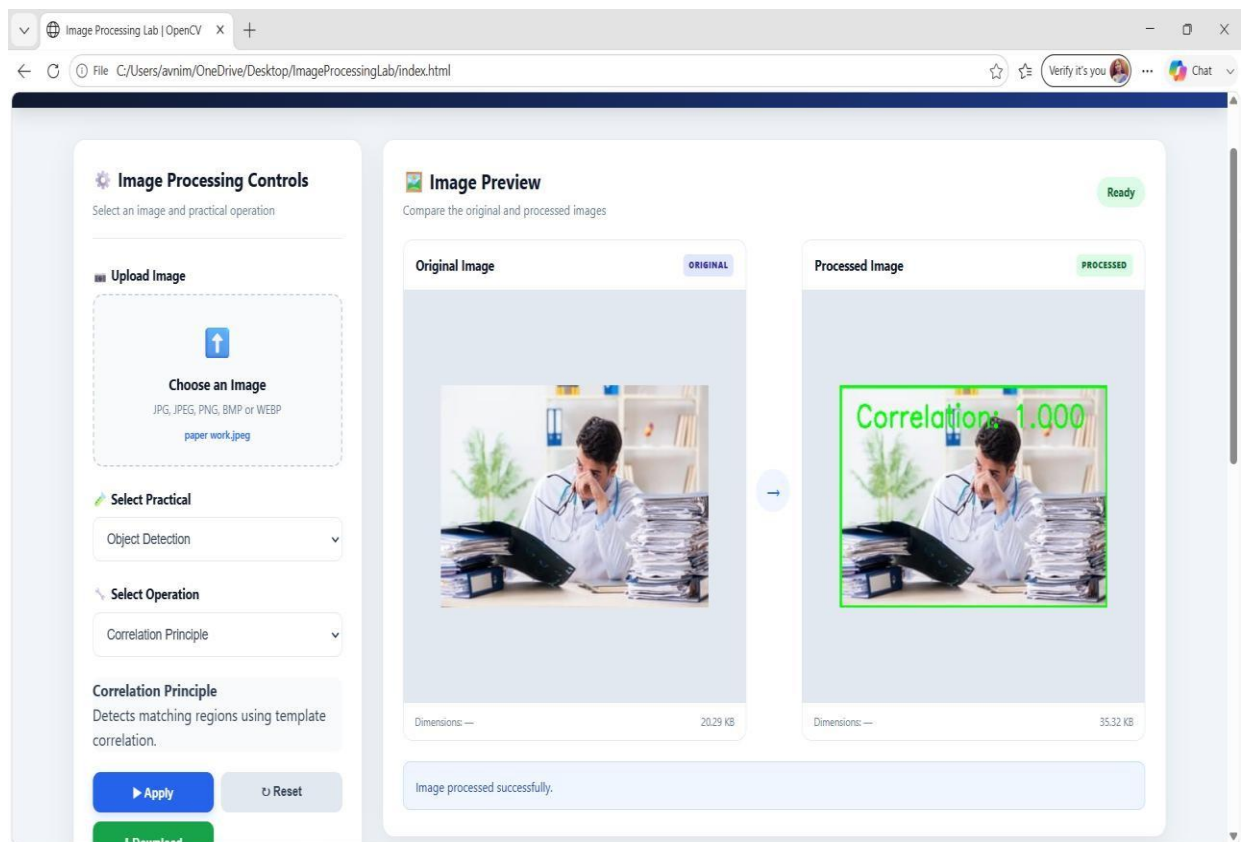
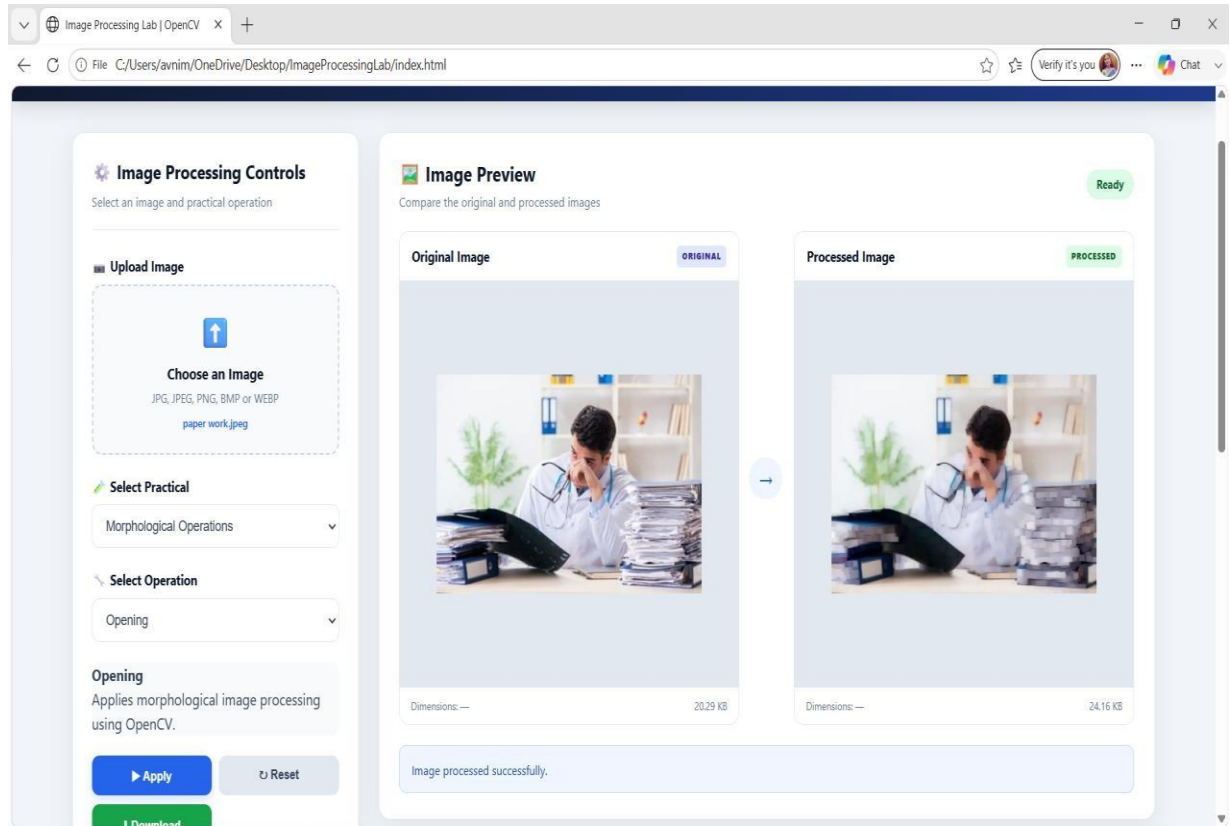


Image Processing Lab | OpenCV

File C:/Users/avnim/OneDrive/Desktop/ImageProcessingLab/index.html

Verify it's you Chat

Image Processing Controls

Select an image and practical operation

Upload Image

Choose an Image

JPG, JPEG, PNG, BMP or WEBP

images (3).jpg

Select Practical

Edge Detection

Select Operation

Sobel

Sobel Edge Detection

Detects edges and boundaries in the input image using the Sobel operator.

Apply

Reset

Download

Image Preview

Compare the original and processed images

Completed

Original Image

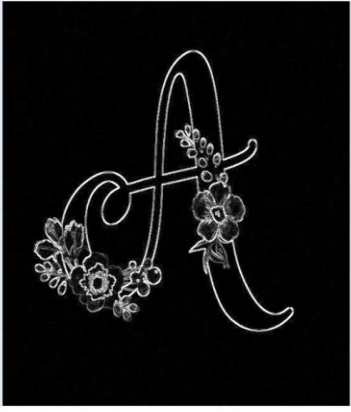
ORIGINAL



Dimensions: — 19.71 KB

Processed Image

PROCESSED



Dimensions: — 72.00 KB

Image processed successfully.

Image Processing Lab | OpenCV

File C:/Users/avnim/OneDrive/Desktop/ImageProcessingLab/index.html

Verify it's you Chat

Image Processing Controls

Select an image and practical operation

Upload Image

Choose an Image

JPG, JPEG, PNG, BMP or WEBP

images (3).jpg

Select Practical

Colour Space Conversion

Select Operation

YCrCb

YCrCb Colour Space

Converts the input image into the YCrCb colour space.

Apply

Reset

Download

Image Preview

Compare the original and processed images

Completed

Original Image

ORIGINAL



Dimensions: — 19.71 KB

Processed Image

PROCESSED



Dimensions: — 38.56 KB

Image processed successfully.

5. CONCLUSION

The Image Processing Lab project successfully demonstrates various fundamental image processing techniques using a web-based interface. The system integrates HTML, CSS, and JavaScript for the frontend with Python, Flask, OpenCV, and NumPy for backend image processing. It provides practical operations such as grayscale conversion, arithmetic and bitwise operations, geometric transformations, image enhancement, spatial filtering, inpainting, compression, morphological operations, and object detection using the correlation principle.

6. REFERENCES

- OpenCV Documentation, “OpenCV: Open Source Computer Vision Library.” Available at: <https://docs.opencv.org/>
- Python Software Foundation, “Python Documentation.” Available at: <https://docs.python.org/3/>
- Flask Documentation, “Flask Web Development Framework.” Available at: <https://flask.palletsprojects.com/>
- NumPy Developers, “NumPy User Guide.” Available at: <https://numpy.org/doc/>
- Gonzalez, R. C., and Woods, R. E., Digital Image Processing, 4th Edition, Pearson Education.